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Paleomagnetic Evidence for the Continuity and Independent Movement of a Distinct Major Crustal Block in the Southern Appalachians

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The magnetization of 22 granitic and gneissic southern Appalachian rock units, with estimated cooling ages of 415–250 m.y., has been determined. Included are 842 samples from 114 sites within 19 granites (100 sites) and 3 gneisses (14 sites) located in North Carolina, South Carolina, and Georgia. Samples from 57 of these sites in 14 granites and 3 gneissic bodies yield a remanent magnetism after af demagnetization which exhibits good within-site and/or between-site directional consistency. Data for units which cooled to temperatures <300°C between 350–240 m.y., a period of apparently only slight North American plate motion, can be divided into two groups. A mean paleopole calculated for the first of these groups, group A (derived from six granites and gneisses 365–325 m.y. in age), located in the vicinity of Atlanta, Ga., is coincident with well-defined Lower Carboniferous North American paleopoles. Site paleolatitude is estimated to be ~11°S. Group B granites (six units) range in age from 350–250 m.y., are located to the SE of an arc drawn from Columbia, S.C., through Athens, Ga., to Macon, Ga., are apparently anomalous, and lie in a crustal block >20,000 km^2 in size. The southeastern extent of this block is unknown due to Coastal Plain onlap. A mean paleopole for this group, with corrections for maximum tilt estimates, exhibits good precision but has a paleosite latitude of ~10°N. Without tilt correction the paleopole for group B still exhibits an anomalous paleosite latitude of ~4°N. These data indicate that the group B block (Elberton-Sparta Crustal Block) lying to the SE has an apparent paleosite latitude corresponding to magnetization at a location to the north of the zone containing group A units. The preferred explanations for these data are (1) a northwest postemplacement tilting of the Elberton-Sparta Crustal Block during the Alleghenian orogenic event in the southeastern Appalachians, or (2) emplacement as a microplate caught between North America and Africa during the collision of these continental masses. Both interpretations appear to be consistent with the available geological, geochemical, and geophysical data for the area. Of those granites which comprise a ~400-m.y.-old group, only one granite, Lowrys, in South Carolina, exhibits good within-site and between-site precision. A paleopole calculated for the sites sampled in this unit exhibits fair precision and is consistent with magnetization acquisition as part of North America in Middle Paleozoic time.

It has been established that southern Appalachian tectonic processes during the Middle Paleozoic–Early Mesozoic were complex, and agreement concerning emplacement mechanism for rocks in this region is not universal. While it has been recognized for many years that parts of the southern Appalachians may be displaced [e.g., *Jonas*, 1932], the timing of these events has not been well established. More specifically, a good understanding of the tectonic effects accompanying Alleghenian (Hercynian) deformation is lacking. A number of models have been proposed, however, to account for the known and inferred structural setting of the southern Appalachians, and these are reviewed elsewhere [e.g., *Hatcher*, 1972; *Griffin*, 1974; *Cook et al.*, 1979].

This study deals with that part of the southern Appalachian orogen represented in Figure 1. The area has been divided into a series of tectonic provinces or belts [*King*, 1955], the geology of which has been summarized elsewhere [e.g., *Hatcher*, 1972]. Granites and gneisses sampled for this study are exposed in the Blue Ridge, Inner Piedmont, Charlotte Belt, Carolina Slate Belt, and Kiokee Belt of Georgia and North and South Carolina, and are labeled with two letters in Figure 1. Unit names are listed with abbreviations, state locations, and ages in Table 1. These include 114 sites totaling 842 samples from 3 granites in North Carolina, 5 granites and 1 gneiss in South Carolina, and 11 granites and 2 gneissic units in Georgia. It was the purpose of this study to present the paleomagnetic evidence from selected granites and gneisses of Mid-Late Paleozoic age in Georgia and North and South Carolina and to relate these data to the available geophysical and geochemical information in the region. When samples were originally obtained from the granites in North and South Carolina, it was assumed that ages were 350 m.y. or less. Some of these granites, however, have been dated at ~400 m.y. [e.g., *Butler and Fullager*, 1978]. Paleomagnetic data for these older granites are also reported here.

It must be pointed out that while significant thrust events yielding displacements of >50 km have been suggested for the Inner Piedmont, no direct age control exists to pinpoint the timing of such events. Timing has been inferred, however, by extrapolating apparent Late Paleozoic thrusting in the Valley and Ridge and Blue Ridge provinces across the Inner Piedmont and into the Charlotte-Slate belts to the southeast. In this report, an effort is made to correlate paleomagnetic data from southeastern Appalachian Piedmont Mid-Late Paleozoic intrusives to much of the already reported geological, geochemical, and geophysical data. These results should help to place some limits (temporal and spacial) on Mid-Late Paleozoic tectonic events in the region.

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PREVIOUS WORK

Regional Paleomagnetic Data

Only a few paleomagnetic results for the southeastern Appalachians have been presented. Most of these are in abstracts presented at national or regional GSA and AGU

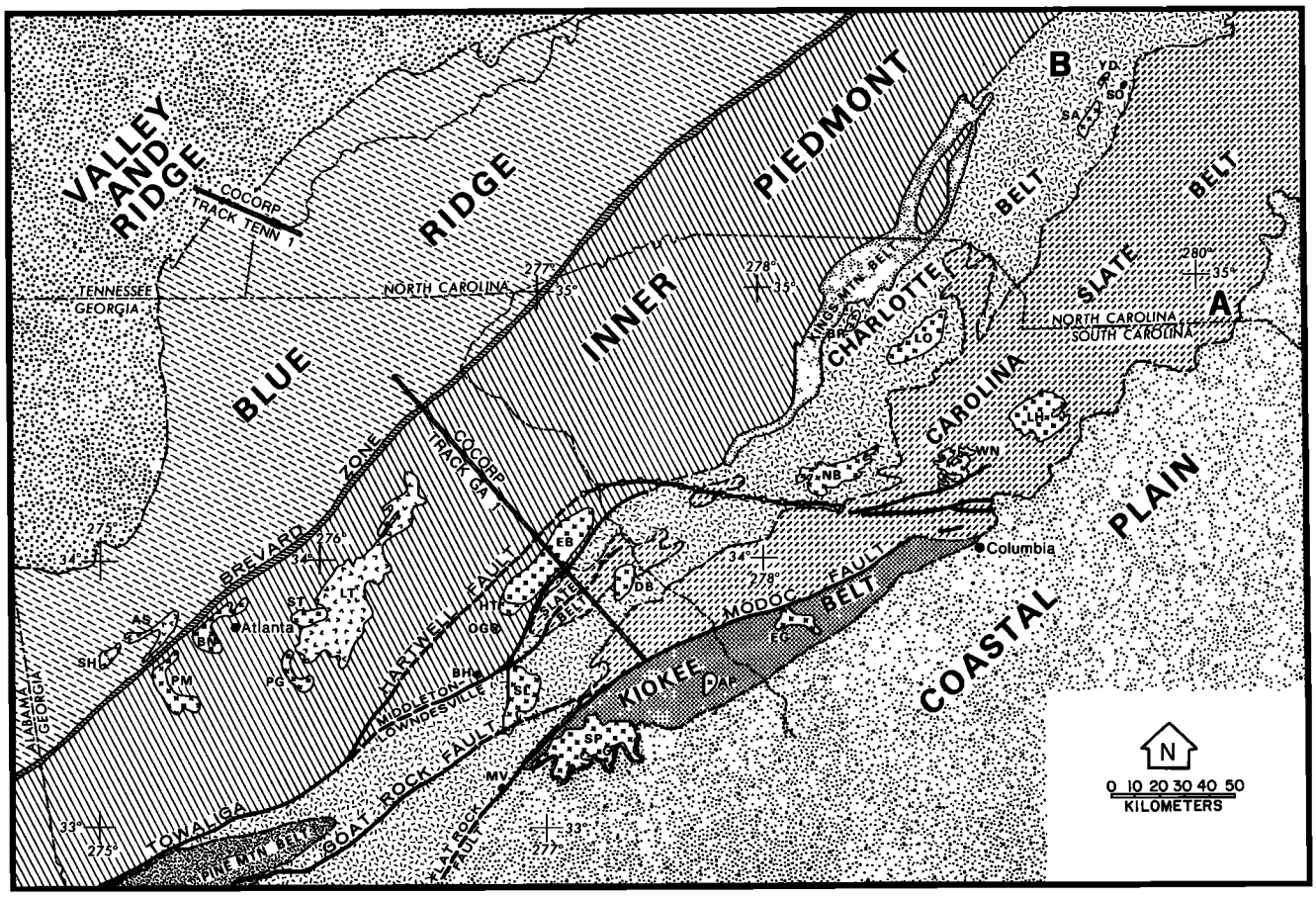


Fig. 1. Study area in the southeastern Appalachian Piedmont. Letters denote units sampled. Details are given in Table 1. Granites OG and HT and the LT gneiss were sampled but not used due to poor age control.

meetings [e.g., *Brown and Barton,* 1980; *Brown,* 1981], although a few reviewed papers have been published [e.g., *Ellwood and Whitney,* 1980; *Ellwood et al.,* 1980; *Ellwood and Wenner,* 1981; *Ellwood,* 1981; *Ellwood and Abrams,* 1982]. These papers have concentrated on the magnetization of two major units in the southeast, the Elberton granite and Austell gneiss. This work has indicated that no major structural reorientation of either body (tilting >35°) is necessary to produce a fit between the paleomagnetic poles determined for these units and published apparent polar wander paths for North America [e.g., *Irving,* 1979; *Van der Voo,* 1979] although small tilt magnitudes are indicated. Thrusting of these units is not excluded by the paleomagnetic data, and *Ellwood and Abrams* [1982] have even indicated that large-scale left-lateral strike-slip translation, followed by thrusting to the NW, is one possible way to explain the data from the Austell gneiss. However, this is not their preferred explanation for the data. Since in situ cooling at or near the present locations of these units is the simplest explanation for the Elberton and Austell data, the genetic models developed by these authors are based on in situ magnetization with slight tectonic tilting (<35°) and later uplift [*Ellwood et al.,* 1980; *Ellwood and Abrams,* 1982].

Regional Gravity Data

A compilation and interpretation of gravity data covering the southern Appalachians in Georgia and South and North Carolina has been made by *Long* [1979]. These same data are used in compiling the Appalachian gravity map of *Haworth et al.* [1980].

Long [1979] has suggested that regions characterized by negative anomalies that are coincident with the Inner Piedmont imply an overcompensation at the base of the crust by the existence of low-density continental crustal-type rocks, whereas the areas of positive anomalies are characterized by higher density, more mafic lithologies. Thus the regional gravity gradient marks the boundary between two major crustal blocks that differ in thickness and/or composition.

Modeling of the Bouguer gravity data by *Long* [1979] indicates that the contrasting lithologies equated with the regional gravity gradient probably extend to considerable depth (>30 km). In contrast, modeling within the Charlotte-Carolina Slate-Kiokee belts, in the region characterized by positive anomalies, suggests that high-density, mafic lithologies exist down to ~20 km beneath the present-day erosional surface [*Long,* 1979]. Further, *Long* [1979] has suggested that at least one large fragment of rifted continental crust is indicated for this region.

Although the depth of magma genesis is uncertain for the late-orogenic granites, rough estimates suggest that many of the granites must have originally formed through anatectic processes that occurred approximately 12 km or more below the present-day erosional surface [*Whitney et al.,* 1976]. Thus it appears that the lithologies inferred from gravity modeling reflect the contrasting nature of the rock type from which the late-orogenic granitoids were derived, namely,

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TABLE 1. Unit Names and Ages

Notation	State	Unit Name	N	Age	Reference
AP	GA	Appling granite	18	late Paleozoic	*Fullagar and Butler* [1979]
AS	GA	Austell gneiss	63	(365–325)	*Ellwood and Abrams* [1981]
BH	GA	Buckhead granite	10	(350–320)	this paper
BN	GA	Ben Hill granite	24	~325	*Atkins et al.* [1981]
BR	SC	Bald Rock granite	18	388 ± 5 minimum	P. Fullagar, personal communication, 1981
DB	GA	Danburg granite	12	late Paleozoic	*Fullagar and Butler*, 1979
EB	GA	Elberton granite	291	(350–320)	*Ellwood et al.*, 1980 *Ross and Bickford*, 1980
EG	SC	Edgefield granite-gneiss	12	(317–254)	*Snoke et al.* [1980*a*, *b*]; R. D. Dallmeyer, personal communication, 1982
LH	SC	Liberty Hill granite	30	293 ± 15	*Fullagar and Butler* [1979]
LO	SC	Lowrys granite	19	399 ± 4	*Fullagar* [1971]
MV	GA	Milledgeville granite	6	(350–320)	this paper
NB	SC	Newberry granite	6	415 ± 9	P. Fullagar, personal communication, 1981
PG	GA	Panola granite	36	~325	*Atkins and Higgins* [1980]
PM	GA	Palmetto granite	36	~325	*Atkins et al.* [1981]
SA	NC	Salisbury granite	30	402 ± 2	*Butler and Fullagar* [1978]
SH	GA	Sand Hill gneiss	12	(365–325)	*Ellwood and Abrams* [1982]; this paper
SL	GA	Siloam granite	48	281 ± 3	*Fullagar and Butler* [1979]
SO	NC	Southmont granite	6	386 ± 8	*Butler and Fullagar* [1978]
SP	GA	Sparta granite	81	289 ± 2	*Fullagar and Butler* [1979]
ST	GA	Stone Mountain granite	48	(325–291)	*Whitney et al.* [1976]; *Atkins and Higgins* [1980]
WN	SC	Winnsboro granite	30	295 ± 4	*Fullagar and Butler* [1979]
YD	NC	Yadkin granite	6	413 ± 4	*Butler and Fullagar* [1978]

N = number of samples from each granite; notation is from locations in Figure 1. The age for the Sand Hill gneiss is based on close proximity to the Austell gneiss and the fact that both units have experienced the same deformational history.

from continental crustal rocks for plutons that lie within the Inner Piedmont and from more variable but generally more mafic lithologies in most regions of the Charlotte-Carolina Slate-Kiokee belts.

Regional Magnetic Anomaly Data

A compilation of much of the available magnetic anomaly data covering the Appalachian orogen has been published by *Zietz et al.* [1980]. A distinct pattern exists of less variable (long wavelength) anomalies to the NW of the Towaliga-Hartwell Fault (Figure 1) versus highly variable (short wavelength) anomalies immediately to the SE of this line. The Charlotte-Carolina Slate belts are characterized by generally low magnetic anomalies with scattered and local high magnetic values. These highs can often be directly correlated with relatively young near-surface mafic intrusives, while the general low values are the result of metamorphism which has affected the older rocks in the area.

The magnetic anomaly data is significant because it is similar to the gravity data in two ways. First, both data sets exhibit low wavelength anomalies to the NW of the Charlotte belt and high wavelength anomalies to the SE of the Inner Piedmont. And second, the change from low to high wavelength anomalies occurs in both data sets at the boundary between the Inner Piedmont and Charlotte belt. These data indicate a distinct difference between the two regions, not only at the surface, but in the near surface and at depth in the crust.

Seismic Data

A COCORP seismic reflection profile designed to look at deep crustal and subcrustal reflectors [see *Cook et al.*, 1979] was conducted, starting in the Valley and Ridge province of Tennessee and ending at the Modoc line in Georgia (Figure 1). Three gaps exist in the data set, a large gap which includes the high Blue Ridge between Murphy, North Carolina, and Helen, Georgia, and two gaps in the Inner Piedmont. One of these includes the northwestern margin of the Elberton granite where the line encountered the Broad River and some large granite quarries immediately to the SE of the river. Reflectors above 1–2 s (3–6 km) in the Piedmont are generally not continuous due to data acquisition problems. The COCORP line was later extended to cross the Coastal Plain of Georgia [*Cook et al.*, 1981].

Cook et al. [1979] have inferred from their reflection data (COCORP Tennessee line 1, Figure 2) that a major, nearly horizontal thrust fault exists at shallow crustal depths in Tennessee and continues below the Piedmont and under the Coastal Plain. This is based on interpreting the relatively horizontal reflectors in Tennessee as a layered sequence of Lower Paleozoic sedimentary rocks underlying overthrust younger rocks. They suggest that such a pattern generally exists across the Piedmont to the Modoc line. While a similar pattern appears to exist below the northwestern Inner Piedmont surface, no direct seismic correlation has been presented to that effect, since the high Blue Ridge is missing from the COCORP traverse [*Cook et al.*, 1979]. Further, this pattern does not appear to exist in the vicinity of the

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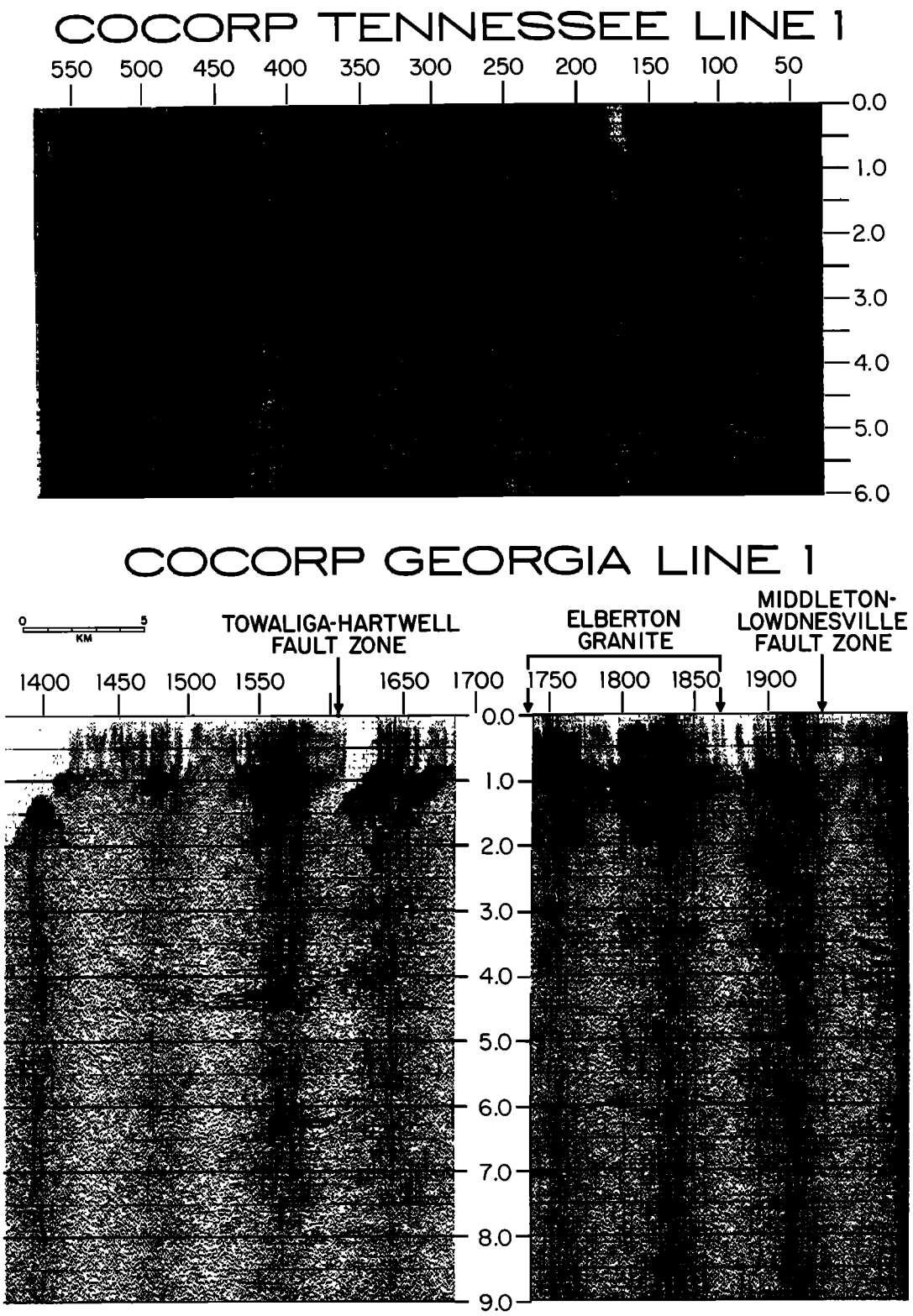


Fig. 2. Segments of the COCORP Georgia Line 1 [*Cook et al.*, 1979] with location of the Elberton granite and the Middleton-Lowndesville fault zone given. Segment of the COCORP Tennessee Line 1 given for comparison.

Elberton pluton (COCORP Georgia line 1, Figure 2), nor does it exist to the southeast of the Elberton, at least as far south as the Modoc line. Within the Charlotte and Carolina Slate belts the seismic reflection pattern is more varied and complex than to the NW.

It is interesting to note that the deep seismic character of the crust along the COCORP line is significantly different to the SE of the Inner Piedmont as compared to the NW of the Charlotte belt. An abrupt change occurs at the Inner Piedmont-Charlotte belt boundary (near the Elberton granite), the point at which the gravity and magnetic anomaly signatures change. This change has been interpreted by *Cook and*

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Oliver [1980] as a point of transition from Grenville continental crust in the northwest to oceanic crust or attenuated continental crust to the southeast of the Inner Piedmont.

The geologic map of the southern Appalachian region by *Williams* [1978] indicates that the Elberton granite may possibly be rooted. From this it might be inferred that the unit extends (extended) to a relatively great depth. To test if the Elberton is or is not rooted, the COCORP Georgia traverse was directed across the body. *Cook et al.* [1979] have demonstrated that below the Elberton at depths >6 km, reflectors are relatively continuous. They inferred from these data that the Elberton granite is rootless and therefore has been transported to its present location.

Mapping of the Elberton has shown that the granite is an extremely uniform, fine grained body which has an areal distribution of >500 km^2 [*Ellwood and Whitney*, 1980]. The fine grained nature of the unit has been explained from analysis of geochemical data, indicating that the magma, which originated at 18–20 km depth, rose isothermally to a point at which crystallization and vapor generation sharply increased the viscosity of the granite (at 13–15 km depth), preventing further rise [*Stormer et al.*, 1980]. *Stormer et al.* [1980] suggested that this crystallization process produced a sill-like shape only a few kilometers thick. Other geological evidence supports the idea of a relatively thin sill-like (or even dike-like) body. For example, a breccia pipe (such pipes are known to occur as intrusive fluidized systems) [e.g., *Reynolds*, 1954] 10 m in diameter was recently discovered in the Elberton granite. Incorporated within this pipe are xenolith fragments which are only slightly rounded. Partially entrained wall rock fragments exhibit arcuate peels which are convex downward. These data indicate upward migration (indicated by peel orientations) of the material in the pipe over fairly short distances (indicated by the lack of substantial rounding in xenoliths), with the origin of the pipe at or near the lower contact of the Elberton with the country rock (indicated by the presence of nongranitic xenoliths).

If the Elberton granite is relatively thin, as these observations indicate, then the seismic data so far presented are insufficient to establish whether or not the granite is allochthonous. This uncertainty must be considered when evaluating the timing of thrust events in the Inner Piedmont.

Further information on the timing of the Piedmont thrust event has been presented by *Harris et al.* [1981]. By using U.S. Geological Survey seismic data from Tennessee and North Carolina and the data of *Cook et al.* [1979], *Harris et al.* [1981] reinterpret the data of *Cook et al.* [1979] to indicate that 'the horizontal events identified by Cook et al. as Paleozoic sedimentary rock in the subsurface beneath the Piedmont Allochthon are in fact Precambrian metasedimentary and volcanic rocks similar to those exposed in the Fries thrust sheet.'

Oxygen Isotopes

$^{18}O/^{16}O$ data have been obtained [*Wenner et al.*, 1977; *Wenner*, 1981] from a number of late-orogenic plutons from the Piedmont province of Georgia and South Carolina and portions of North Carolina. These data appear to reflect contrasting protoliths, in which the ^{18}O-enriched plutons formed through some anatectic process from metasedimentary, sialic protoliths, whereas the low-^{18}O granites within the Charlotte-Carolina Slate and portions of the Kiokee belts were derived from source regions dominated by more mafic, igneous rocks. Thus it would appear that this series of granitoids exhibits a number of isotopic similarities to the S and I type granitoids first described in Australia [see *O'Neil and Chappell*, 1977]. Such an interpretation is supported by the correlative initial strontium isotopic data [*Fullagar and Butler*, 1979], by the striking contrast in the nature of the exposed geologic terranes into which the plutons are intruded, by the gravity data which suggest contrasting deep crustal lithologies, and by the magnetic anomaly data, which indicate gross differences in crustal lithologies. Furthermore, this isotopic pattern clearly implies that any major tectonic displacement (>50 km) that may have affected the Piedmont must have occurred prior to intrusion of these granites. A maximum limit of 50 km of possible offset in these terranes is based on the location of the Stone Mountain granite (ST in Figure 1), ~50 km to the NW of the Inner Piedmont-Charlotte belt boundary, and the assumption that high $\delta^{18}O$ values exhibited by this granite were derived from deep crustal, Inner Piedmont lithologies. The Elberton granite, however, which exhibits transitional $\delta^{18}O$ values lies at the apparent transition in deep crustal lithologies from continental crust to more mafic types. While this may be a geological accident, if the overall $\delta^{18}O$ granite values are combined with the Elberton results, the simplest explanation for the data is zero or very small magnitude (<10 km) thrusting since emplacement of the Elberton granite.

Magnetic Susceptibility

The magnetic susceptibility in seventeen of the granites for which oxygen isotopes have been determined also has been evaluated [*Ellwood and Wenner*, 1981]. These data indicate that there is an inverse relationship between susceptibility and $\delta^{18}O$. Since it has been shown that magnetite is depleted in granites derived from carbonaceous rocks (equivalent to S-type or crustal-derived granites) and enriched when derived from low carbonaceous rocks (equivalent to I-type or more mafic derivative rocks) [*Ishihara*, 1977], and, since susceptibility can be related directly with magnetite content (summarized, for example, by *Nettleton* [1976]), this magnetite-$\delta^{18}O$ relationship is not too surprising. Of the six granites with lowest susceptibility values (low magnetite–crustal derived granites), five (BN, EB, PG, PM, and ST in Figure 1) are located in the Inner Piedmont at or on the boundary of the Piedmont gravity low. All but one of the granites to the SE of the Inner Piedmont have relatively high susceptibilities (high magnetite–more mafic derivative rocks). These data further substantiate the conclusion reached from the oxygen isotopic data, namely, that any major Inner Piedmont tectonic displacement (>50 km) occurred prior to emplacement of the grantites.

METHODS

Field

A total of 842 cores were obtained by using a gasoline-powered portable drill from 114 sites in 19 different granites and 3 gneisses (Figure 1; Table 1). Sites chosen were quarries, road cuts, and a few flat-lying outcrops. Orientation of cores was made by using a Brunton compass with some sun compass readings. Core sites were selected which avoided flame cut surfaces, high weathering zones, pegmatites, xenoliths, and shear zones. Lateral separation between cores averaged >5 m.

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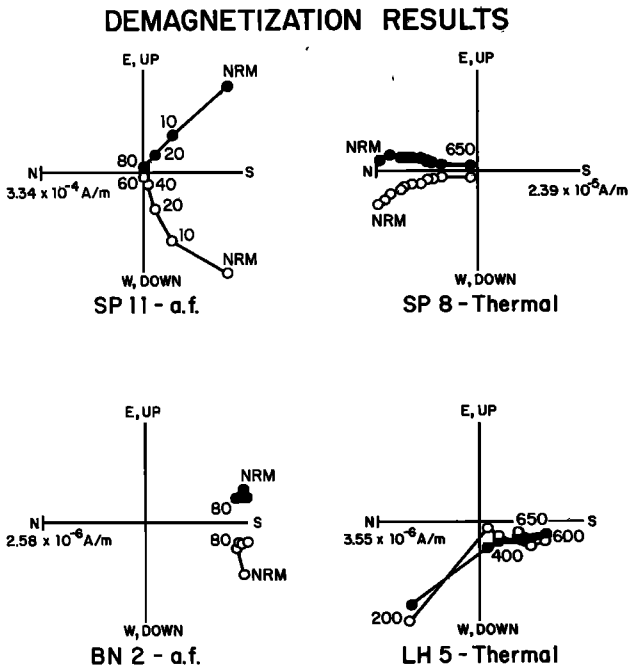


Fig. 3. *Zijderveld* [1967] vector plot of af and thermal demagnetization data for sample SP 11, BN 2, SP 8, and LH 5. Solid symbols are horizontal projection; open symbols are vertical projection.

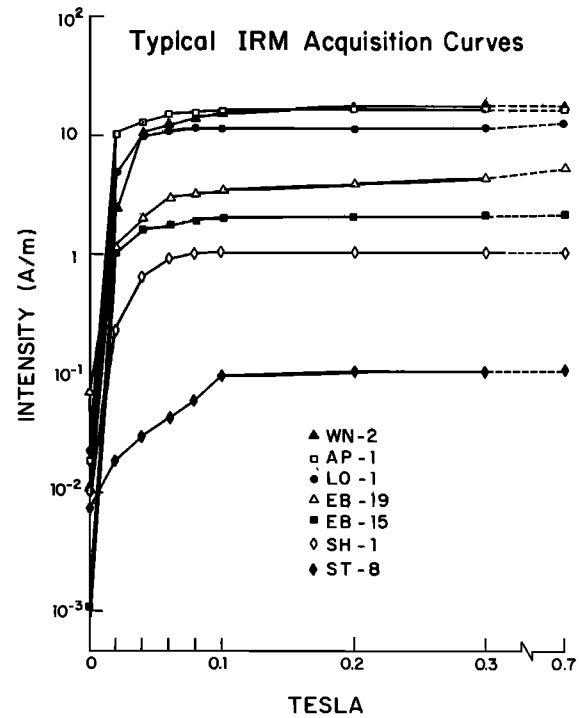


Fig. 4. Typical isothermal remanent magnetization (IRM) acquisition plots of intensity of magnetization (A/m) against dc magnetic field (tesla).

Laboratory

The cores of ~2.5-cm diameter were sliced to lengths of ~2.2 cm to conform with the established optimum length to diameter ratio range of 0.82–0.90 [*Banerjee and Stacey*, 1967; *Noltimier*, 1971]. The natural remanent magnetization (NRM) was measured for all samples by using a Schonstedt SSM1-A spinner magnetometer coupled with a SWTP 6800 microprocessor.

To isolate and identify the characteristic or primary magnetization, a stepwise af demagnetization was performed on pilot samples from all sites at 10 mT or 20 mT steps to 90 mT. Typical results are plotted in Figure 3.

A selected suite of samples from each unit was thermally demagnetized at 100°C intervals by using a Schonstedt thermal demagnetizer. Thermal as well as af demagnetization was used in an attempt to isolate spurious components of magnetization [*Parry*, 1967].

A sample from each lithologic and textural variation from this data set was selected for microscopic analysis. After RM measurement a polished section was prepared from this sample and examined under reflected light. This sample was given an isothermal remanent magnetization (IRM) in fields of 20, 40, 60, 80, 100, 200, 300, and 700 mT and the magnetization measured. Typical results are plotted in Figure 4. The same samples were then af demagnetized at 10 mT steps to 90 mT and the RM measured after each step. Typical data are presented in Figure 5.

RESULTS

Vector plots for typical granite samples (Figure 3) illustrate the wide range of magnetic behavior observed for these units. Even within the same unit, extreme variability exists, probably as a result of the large range of coercivities observed for such units [*Ellwood et al.*, 1980]. Often, a single specimen may appear to be relatively unaffected by af or thermal demagnetization (Figure 3), while other samples (often from the same site) may exhibit more ideal behavior with demagnetization (decay to the origin typical of a characteristic magnetization as discussed, for example, by *Zijderveld* [1967]; Figure 3). Within single units, however, there may be very little correspondence in directions for such samples.

The magnetic carrier ranges from magnetite to hematite and appears generally to be very fine grained in nature. Both

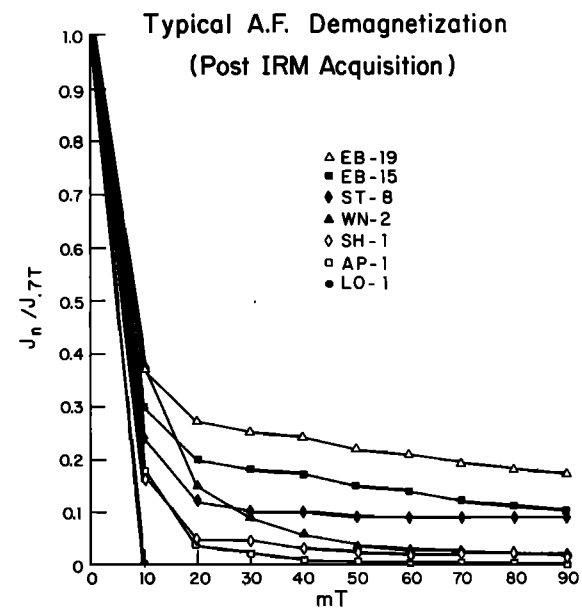


Fig. 5. Normalized intensity of magnetization (Jn/0.7 T) against af demagnetization field (mT) of samples in Figure 8 after IRM acquisition in a field of 0.7 T.

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TABLE 2. Site Mean Remanent Magnetic Results

Site	N1/N2	D	I	$J \times 10^{-2}$	κ	$\bar{\chi} \times 10^{-5}$
(a) 250 to 365 m.y. old Granites and Gneisses						
AP 1	6/6	172.4	−5.2	24.2	1.9	48.7
AP 2	6/6	276.2	−12.7	109.0	2.7	39.1
AP 3*	6/4	309.0	34.5	19.9	44.9	35.4
BH 1*	10/10	157.1	−1.9	7.5	179.1	0.3
BN 1*	6/6	161.9	27.7	0.7	124.8	1.4
BN 2*	6/3	165.6	13.4	3.3	871.3	2.3
BN 3*	6/6	154.6	9.1	1.1	55.6	1.4
BN 4*	6/3	171.0	19.0	2.6	19.3	1.0
DB 1	6/5	102.0	37.8	3.9	2.7	69.1
DB 2	6/6	168.2	66.6	3.6	2.1	103.0
EB 22*	6/6	334.5	13.2	0.4	16.2	1.3
EB 23	7/7	44.7	62.4	0.3	2.8	0.9
EB 24*	6/6	180.7	2.3	4.3	8.7	0.8
EB 25*	6/6	166.6	14.7	39.6	29.4	1.2
EG 1*	6/6	169.4	−5.8	2.3	48.5	1.4
EG 2*	6/6	173.0	−11.7	3.6	71.9	0.6
LH 1	6/6	86.5	57.7	0.9	1.8	16.7
LH 2	6/6	77.6	59.5	0.3	3.5	18.7
LH 3*	6/6	33.4	72.3	0.7	6.4	13.4
LH 4	6/6	187.1	86.2	0.4	1.7	16.8
LH 5	6/6	110.5	50.0	0.3	3.6	39.8
MV 1*	6/6	164.4	0.4	2.4	206.3	1.1
PG 1*	6/6	166.8	11.1	2.1	22.1	2.8
PG 2*	6/4	177.4	−0.8	0.8	10.9	2.4
PG 3*	6/5	185.4	9.5	5.6	7.7	2.2
PG 4	6/6	50.8	32.3	5.7	1.4	1.7
PG 5	6/6	141.1	33.8	2.8	1.3	0.5
PG 6	6/6	288.4	−0.3	6.6	1.7	0.3
PM 1*	6/4	126.0	20.1	4.0	33.1	3.7
PM 2	6/6	228.3	−15.0	35.4	2.8	7.6
PM 3*	6/3	153.0	14.2	34.7	17.6	2.8
PM 5	6/6	310.5	−28.5	2.6	2.8	11.4
PM 6*	6/4	151.2	8.7	12.4	15.2	7.9
SH 1*	6/4	139.7	15.5	0.5	30.9	0.9
SH 2*	6/3	152.6	37.4	0.1	9.8	0.3
SP 1	8/8	105.2	32.4	0.4	1.9	14.6
SP 2	6/6	305.0	51.7	1.4	2.5	45.0
SP 3	6/6	204.8	67.3	1.2	1.7	9.3
SP 4	6/6	50.3	69.4	2.7	2.2	41.7
SP 5	6/6	189.5	36.6	9.9	1.8	32.1
SP 6*	6/6	319.4	25.9	11.7	8.9	34.5
SP 7*	6/6	167.8	−29.3	34.7	226.4	11.9
SP 8	6/6	94.2	28.6	40.4	1.3	34.8
SP 9	6/6	209.7	45.9	56.6	1.4	36.1
SP 10*†	7/7	153.4	−6.9	2.7	60.7	28.1
SP 11	6/6	112.8	−12.5	67.6	1.2	31.7
SP 12	6/6	207.2	59.6	110.0	1.2	38.6
SP 13	6/6	350.5	31.0	2.1	4.8	36.6
SL 1	6/6	260.9	48.4	86.1	1.2	24.0
SL 2	6/6	13.3	19.9	63.4	1.2	19.5
SL 3	6/6	272.5	33.8	1.8	1.7	16.4
SL 4	6/6	54.5	15.1	5.4	1.6	27.0
SL 5	6/6	1.3	7.9	1.0	0.9	39.9
SL 6	6/6	316.7	52.0	7.0	2.1	18.0
SL 7	6/6	303.1	−54.9	61.8	2.0	27.6
SL 8*	6/6	331.5	16.9	18.3	8.4	42.2
ST 1*	6/4	10.1	21.7	0.1	50.6	0.3
ST 2*	6/5	243.5	26.7	0.1	138.4	0.2
ST 3*	6/4	127.0	30.8	0.1	13.3	0.2
ST 4*	6/6	131.3	31.2	0.1	49.7	0.2
ST 5*	6/4	325.3	32.6	0.1	98.7	0.2
ST 6*	6/6	138.7	12.4	0.1	13.5	0.1
ST 7*	6/6	89.1	11.7	0.1	12.7	0.6
ST 8	6/6	98.5	20.5	0.1	4.9	0.6
WN 1	6/6	212.5	64.0	0.6	2.0	2.3
WN 2	6/6	0.7	63.8	0.8	3.3	1.8
WN 3	6/6	76.3	45.1	0.4	2.1	1.9
WN 5	6/6	200.4	53.0	0.5	2.2	1.8
WN 6	6/6	167.6	67.5	1.0	4.2	4.0
(b) 385 to 415-m.y.-old Granites						
BR 1	6/6	271.5	38.8	8.2	2.1	12.3
BR 2	6/5	219.4	−4.7	6.9	1.1	17.4
BR 3	6/6	283.7	41.8	10.2	2.2	32.5
LO 1*	6/5	179.9	33.4	0.1	6.3	0.4
LO 2	6/6	33.9	57.9	18.1	1.7	57.8
LO 3*	7/6	154.3	19.2	8.6	23.1	11.3
NB 1	6/6	57.4	39.2	0.9	2.1	40.3
SA 1*	6/5	29.6	41.1	1.7	52.8	26.6
SA 2	6/6	134.9	45.7	0.5	1.6	16.8
SA 3	6/6	237.0	4.0	4.8	1.3	11.3
SA 4	6/6	212.3	47.2	0.6	1.9	17.7
SA 5	6/6	309.8	65.1	0.5	1.5	15.8
SO 1	6/6	268.3	38.0	0.8	2.2	18.9
YD 1*	6/4	4.3	53.5	1.6	247.6	13.2

Site = site name; unit name is given in Table 1. *N*1 = number of samples drilled at the site; N2 = number of samples used in calculating the site mean remanent magnetic directions; *D* = mean declination; *I* = mean inclination; *J* = intensity of magnetization, A/m; $\bar{\chi}$ = bulk susceptibility in gauss/oersted. All RM data are after af demagnetization to 40 or 60 mT.

*Fair to good within-site precision ($\kappa > 5.0$, where κ is an estimate of precision—from *Fisher* [1953]).

†Gabbro associated with a granite.

magnetite and some hematite are apparent from IRM (Figure 4) and af demagnetization after IRM acquisition (Figure 5) results. For example, samples EB-15 (grey) and EB-19 (discolored pink), both from the Elberton granite, exhibit saturation and near saturation, respectively, in fields of 0.2–0.3 tesla, characteristic of magnetite, while EB-19 does not fully saturate in a field of 0.7 T; that result and the red discoloration indicate the presence of hematite (Figure 4). With demagnetization, the IRM is initially rapidly reduced, but then generally exhibits a stronger resistance to higher demagnetization fields than do samples, such as LO-1 (Figure 5) from the Lowrys granite, which have, based on optical examination, abundant coarse-grained magnetite.

Thermomagnetic measurements on pilot samples show high Curie points (Tc > 400°C) indicative of magnetite, hematite, and mixtures of the two as fine grained assemblages [*Davison*, 1982]. Also reported are Curie points of 564°–580°C (magnetite) for Elberton granite samples and blocking temperatures for Austell gneiss samples of >500°C.

Based on the demagnetization data for pilot samples, and on the previous work reported for some of these units (EB, *Ellwood, et al.* [1980], MV, *Ellwood* [1981], and AS, *Ellwood and Abrams* [1982], all samples were af demagnetized at 20 mT steps to 80 mT after an initial NRM measurement. Site mean directions after af demagnetization to 40 or 60 mT are given in Table 2. These af fields were chosen to avoid viscous magnetic components observed for some samples at lower fields. Those samples with fair to good within-site precision ($\kappa > 5.0$ (asterisk) in Table 2) which are 365–250 m.y. have been used to determine unit means (Figure 6), and these data are reported in Table 3. In the case of the Stone Mountain granite (ST), the mean exhibits poor precision ($\kappa = 1.7$). Therefore, anomalous site mean directions were rejected in calculating the Stone Mountain mean direction (Table 3). Also given in Table 3 and plotted in Figure 6 are the unit mean directions for the Austell gneiss [*Ellwood and Abrams*, 1982] and Elberton granite [*Ellwood et al.*, 1980].

Although most of the directions determined for these bodies yield southern hemisphere paleomagnetic poles, the antipodal poles (northern hemisphere poles) are plotted in

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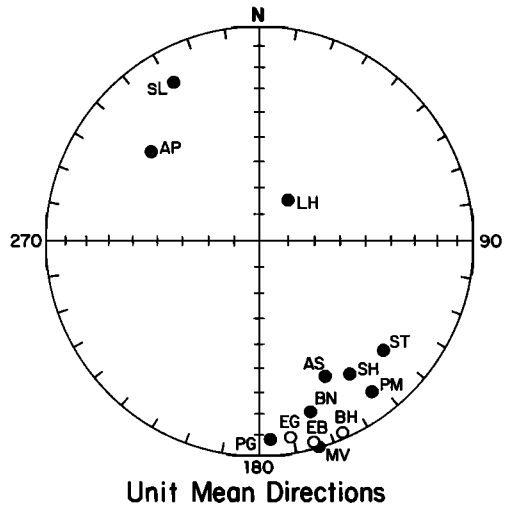


Fig. 6. Unit mean directions for the granites and gneissis (365- to 250-m.y.-old) for sites which exhibit good precision.

Figure 7. Further, where used in the text, the antipodal pole is also reported to facilitate comparison with other published work.

TABLE 3. Individual Stable Site Remanent Magnetism Mean Results

Unit	*N3/N2*	*D*	*I*	κ	θ	ϕ
AP*	1/4	309.0	34.5	44.9	42.5N	183.0
AS	5/35	161.3	29.6	19.0	37.3S	298.1
BH	1/10	157.1	−1.9	179.1	51.0S	314.9
BN	4/18	163.2	17.4	60.7	44.5S	299.1
EB	23/142	164.8	−1.8	46.7	54.0S	303.5
EG	2/12	171.2	−8.8	276.8	59.6S	295.5
LH*	1/6	33.4	72.3	6.4	58.0N	313.3
LO	2/11	166.3	26.9	18.3	39.1S	296.2
MV	1/6	164.4	0.4	206.3	53.7S	303.5
PG	3/15	176.6	6.7	51.9	53.0S	281.4
PM	3/11	143.7	14.7	27.2	36.5S	322.1
SA*	1/5	29.6	41.1	52.8	61.8N	26.4
SH	2/7	145.5	26.6	21.6	32.0S	315.7
SL	1/6	331.5	16.9	8.4	53.8N	150.1
SP	3/19	153.4	−21.1	21.1	56.9S	330.7
ST	3/16	132.6	24.9	45.5	25.0S	328.2
YD*	6/4	4.3	53.5	247.6	86.2N	30.6

Unit from Table 1 and Figure 1; *N2* = number of samples used; *N3* = number of stable sites; *D*, *I*, and κ as in Table 2; θ = paleopole latitude; ϕ = paleopole longitude. Note that most of these units yield southern paleopole latitudes. Data from the Austell gneiss (AS) are from *Ellwood and Abrams* [1982]. Data from 21 of the 25 Elberton granite (EB) sites are from *Ellwood et al.* [1980].

*Units which yield anomalous directions from only one site and were not used in compiling Figure 7.

DISCUSSION

Time of RM Acquisition

High Curie points and blocking temperatures along with the presence of hematite and relatively titanium pure magnetite indicate that RM acquisition in these units generally occurred at temperatures at least >400°C. (It has been demonstrated that the magnetic effect of secondary maghemite (RM) produced during weathering can generally be removed with af demagnetization at relatively low fields (~10 mT) [*Ellwood*, 1981]). The Inner Piedmont of Georgia (including the Elberton granite) had cooled below such temperatures by 240 m.y. [*Dallmeyer et al.*, 1981]. The Elberton granite was essentially cold by ~200 m.y., when intruding Triassic dikes were quenched against the granite as evidenced by glassy margins in the dikes. Further, *Ellwood and Abrams* [1982] have indicated that the Austell gneiss cooled below 600°C, post 365 m.y. and pre-325 m.y. In summary, the best available information indicates that the time of RM acquisition for the 365- to 250-m.y.-old units in this study is less than 360 m.y. and greater than 240 m.y. (Mid-Late Paleozoic), a time of apparently little relative motion of the North American plate [*Irving*, 1979; *Van der Voo et al.*, 1980].

Remanent Magnetism

It becomes apparent, while working on the magnetic properties of granitic and gneissic units in the southern Appalachians, that there are many problems in trying to get reliable information. Exposures are often very poor and badly weathered. While directions can be extremely scattered within individual units or sites, and precision is generally low, there is a marked clustering of RM directions for these Mid-Late Paleozoic units (Figure 6), which corresponds roughly with directions expected for units of this age [*Van der Voo et al.*, 1980]. Close examination of the southeastern Appalachian directions (Figure 6) reveals some interesting information. Directions from Mid-Late Paleozoic granites and gneisses, located to the northwest of the Hartwell extension of the Towaliga fault in Georgia and around the Atlanta, Ga., area (granites BN, PG, PM, and ST and gneisses AS and SH, Figure 6), although somewhat scattered, generally yield similar paleopole latitudes (Table 4), with a mean paleopole latitude and longitude which falls very close to reported stable North American poles of equivalent age (A in Figure 7). Scatter in the data may in part be the result of slight but unidentified, post granite emplace-

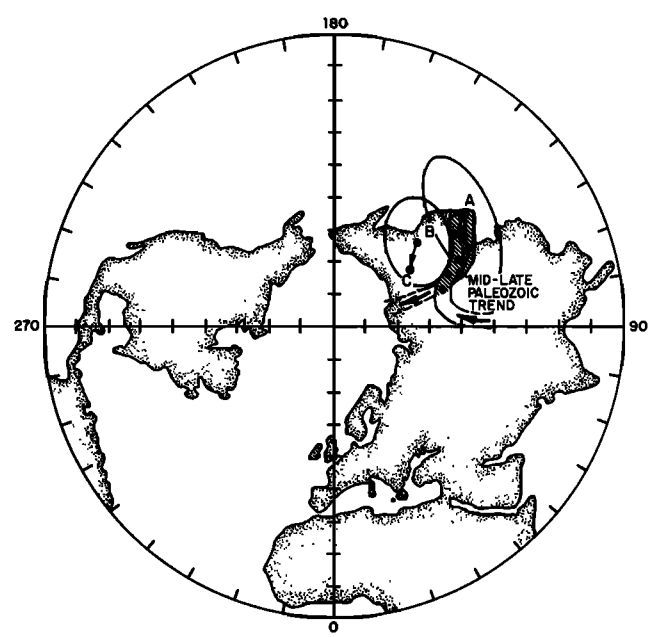


Fig. 7. North polar equal area paleomagnetic pole plot of published North American poles (shaded area) reported by *Van der Voo et al.* [1980] as reliable for the Mid-Late Paleozoic, compared with A, B, and C from Table 4.

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TABLE 4. Paleomagnetic Poles for the Southeastern Piedmont

	N	D	I	κ	θ	φ	dp	dm	Paleosite Latitude
A	6	154.1	20.6	21.9	−39.4	310.2	10.5	17.5	10.6°S ± 8
B	6	160.5	−8.4	48.8	−55.5	313.5	9.9	13.9	4.2°N ± 5
C	6	167.5	−19.7	48.8	−64.0	306.6	12.3	15.4	10.1°N ± 5

A = six units from the Atlanta, Georgia, area (AS, BN, PG, PM, SH, and ST); B = six units from the Elberton Crustal Block (BH, EB EG, MV, SL, and SP); C = the B units from the Elberton Crustal Block after tilt correction (Table 5); N = number of units; D, I, κ, θ, and ϕ as in Tables 2 and 3; dp = the error in determining the great circle distance of the pole from the sample site; dm = an error in declination which corresponds to a displacement from the pole, normal to the pole, sample site great circle. Present-day mean site locations: A = latitude 33.6°N and longitude 276.5°; B, C = latitude 33.5°N and longitude 277.5°. Paleosite latitude = location of the site at the time of magnetization with approximate precision limits based on *Fisher*'s [1953] estimate of precision.

ment block tilting. It has been shown by *MacDonald* [1980] that conventional tilt corrections in such cases may produce declination anomalies in paleomagnetic vectors, and post 250 m.y. brittle deformation in the southeastern Piedmont is known to have occurred [e.g., *Snoke et al.*, 1980*a*]. Poles for granites, gabbros, and gneisses from North and South Carolina, which lie to the north of the dotted line in Figure 1, have been reported which exhibit large scatter and are similar in direction to A and to the North American polar trend [*Brown and Barton*, 1980; *Brown*, 1981; A. Zupan, personal communication, 1980].

Elberton-Sparta Crustal Block

Five granitic units in Georgia and one granite gneiss in South Carolina which exhibit low scatter and similar magnetic characteristics lie in a block to the south and southeast of the dotted line and the line of the Towaliga-Hartwell fault zone in Figure 1. These are labeled EB, BH, MV, SL, SP, and EG in Figure 6. Data for AP (the Appling granite) in Figure 6, for one relatively stable site which appears to be weathered, yield an anomalous direction (Figure 6) and therefore are not considered for paleopole calculations. The EB (Elberton granite) RM data result from analysis of samples from 25 sites within the pluton and are well defined (data for 21 of these sites are given by *Ellwood et al.*, 1980; Table 2). Comparisons between the 320- to 350-m.y.-old Elberton (U/Pb age of *Ross and Bickford* [1980]; Rb/Sr age of *Ellwood et al.* [1980], respectively), MV (Milledgeville granite, one site; *Ellwood* [1980]), and BH (Buckhead granite, one site) paleopole positions (Table 3) show close similarities. The granites also exhibit strong similarities in petrology, grain size, and tectonic style. All these factors suggest an age equivalence for the units. Three other units, the SP (Sparta, three stable sites) and SL (Siloam granites, one stable site) in Georgia and EG (Edgeville granite gneiss, two stable sites) in South Carolina, Mid-Late Paleozoic in age (Table 1), also exhibit paleopole positions which show a close similarity to the Elberton, Milledgeville, and Buckhead positions (Figure 6). These six units are all located in the zone, >20,000 km^2 in size, discussed above (Figure 1), which will be referred to as the Elberton-Sparta Crustal Block. This block is distinguished from segments of the crust to the north and northwest by a unique difference in tectonic style [*Bobyarchick*, 1981], geology, seismic character, magnetic anomaly patterns, gravity, and geochemistry.

Mean paleopoles for granites in the crustal block, without an applied tilt correction (Table 4, B; Figure 7), exhibit steeper than expected paleomagnetic pole latitudes (e.g., pole latitudes for North America during the Mid-Late Paleozoic, reported and summarized by *Kent and Opdyke* [1978], *Kent* [1979], *Van der Voo et al.* [1979], and *Van der Voo et al.* [1980]). Those poles plotted in Figure 7 which are delineated by the shaded zone are from *Van der Voo et al.* [1980]. If a tilt correction (Table 5) suggested by *Ellwood and Whitney* [1980] for the Elberton granite is applied to these data, the paleopole latitude increases (C in Figure 7).

In Table 3 are given the paleosite latitudes calculated for the North American Block data (A) and for the Elberton-Sparta Crustal Block without tilt correction (B) and with tilt correction (C). Two models are preferred for explaining such increases in paleopole latitude and areal consistency in the data. First, en bloc tilting occurred after magnetization of the granites, probably during the Alleghenian (Hercynian) deformational event. Second, the Elberton-Sparta Crustal Block was, at one time, a microplate which was captured during the collision of North America with Africa, which produced the Alleghenian orogenic event. The existence of such a microplate has been discussed by *LeFort and Van der Voo* [1981] (the east Georgia–west Senegal microplate). These two models are discussed below.

Tilt model. Figure 8 is a cartoon which synthesizes the age, geologic, and paleomagnetic information for the Elberton granite in the Elberton-Sparta Crustal Block. Figure 8 represents the inferred tilting history for the entire block. The data indicate that the Elberton intruded at depths of 13–15 km below the surface [*Stormer et al.*, 1980; *Hess*, 1979] between 320 and 350 m.y. [*Ellwood et al.*, 1980; *Ross and Bickford*, 1980], possibly as a sill-like or dike-like body (Figure 8, part 1). After emplacement, the Elberton was tilted down to the northwest, possibly resulting from continental collision, as part of a regionally folded system (Figure 8, part 2). Later, postcollision rebound and uplift of the Inner Piedmont brought the Elberton nearly to the surface at ~200 Ma, tilting it in the process to the SE (Figure 8, part 3). The present-day erosional surface is represented in Figure 8, part 4. This summary of events is predicated on the following information:

Figure 8, part 4 is based on the assumption that the intruding Elberton acquired a magnetic foliation which mirrors the country rock orientations where observed, and this foliation represents the current attitude of the body, inclined at ~30° to the southeast [*Ellwood and Whitney*, 1980].

Figure 8, part 1 is based on the fact that the paleopole calculated for the Elberton is steeper than expected for a unit which acquired an RM magnetization between 350–250 m.y.

TABLE 5. Tilt-Corrected Unit Means

Unit	N3/N2	D	I	θ	φ
BH	1/10	161.7	−16.7	−59.9	314.9
EB	23/142	168.8	−13.5	−60.9	300.4
EG	2/12	177.2	−15.0	−63.8	284.3
MV	1/6	167.2	−10.7	−59.9	302.6
SL	1/6	342.6	29.8	66.5	143.1
SP	3/19	166.6	−32.4	−70.3	317.9

Unit = those units contained within the Elberton Crustal Block; $N3$, $N2$, D, I, ϕ, θ, as in Tables 2, 3, and 4; azimuth of tilt = 100°; dip of tilt = 30°. Assumed azimuth and dip tilt is based on work by *Ellwood and Whitney* [1980].

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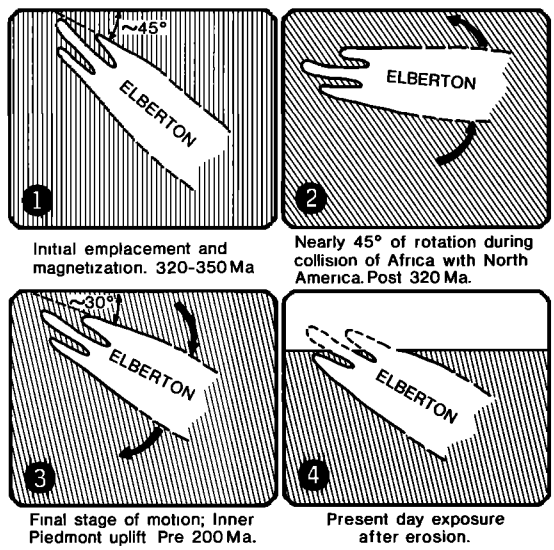


Fig. 8. Cartoon (cross section) illustrating the inferred changes in orientation of the Elberton granite (see text). Pattern orientation equals foliation orientation.

(Figure 7). Therefore, it would be necessary to tilt the block down to the SE by ~15° to bring the pole into coincidence with the North American polar wander path [*Ellwood et al.*, 1980].

Figure 8, part 2 is determined from the assumption that part 1 (Figure 8) is correct, and the following:

1. *Dallmeyer et al.* [1981] have shown that the thermal gradient must have increased in the area of the Elberton between 325 and 275 m.y.

2. The thermal gradient can be increased by creating a thermal blanket at the site.

3. One way to do this is to thrust a rock blanket over the site. Another way to do this is to tilt the block downward through some process such as gentle warping during the slow collision of North America with an external continental block.

4. Such a gentle process is indicated, since the Elberton and most other granites in the block are undeformed.

5. This process has developed the bent or arcuate shapes observed locally (Figure 1) with concave structures directed to the southeast [*Williams*, 1978].

Figure 8, part 3 arises from the following:

1. It has been suggested that vertical en bloc uplift centering on the Inner Piedmont occurred [*Griffin*, 1974] and was very rapid, with the temperature of the Elberton dropping from 500°C or so to <100°C in 75–85 m.y. (275-to ~200 m.y.) [*Dallmeyer*, 1978; *Dallmeyer et al.*, 1981]. This rise may be simple isostatic rebound in response to earlier crustal downwarping. Since at least 13–15 km of uplift and erosion is suggested for the Elberton, it is possible that rocks which were originally thrust over the area have been totally (or nearly totally) removed by erosion from the Inner Piedmont.

2. Such a rebound process brings more mafic (higher density) material, intruded while the block was anomalously deep, to the near surface, and then explains high Bouguer gravity anomalies in the Elberton Crustal Block [*Haworth et al.*, 1980], the extreme magnetic anomaly variability in the Elberton Crustal Block [*Zietz et al.*, 1980], $\delta^{18}O$ values for granites in the Elberton Crustal Block, which are characteristic of a more mafic protolith [*Wenner*, 1981], and some of the brittle deformation observed in the Charlotte-Slate-Kiokee belts (mapping at the University of Georgia; *Snoke et al.* [1980*a*].

It is important to note that each tilt event illustrated in Figure 8 (parts 2, 3) is independently derived by using paleomagnetic data [*Ellwood et al.*, 1980] and isotopic data [*Dallmeyer*, 1978; *Dallmeyer et al.*, 1981], respectively. In summary, the geochemical, geological, and geophysical data from the Elberton-Sparta Crustal Block are internally consistent and indicate two episodes of slight magnitude tilting between 320 and 200 m.y.

Microplate model. The coherent but distinctive pattern exhibited by much of the geological, geophysical, and geochemical data from the Elberton-Sparta Crustal Block demonstrate an en bloc continuity that is considerably different from the bounding crust to the north and northwest. This coherent difference may indicate that the Elberton-Sparta block was once a microplate which has been accreted to North America. Paleosite latitude determined for the block indicates that granite magnetization occurred at 4–10°N latitude. Since both dextral and sinistral strike slip motion has been documented for the Appalachians (summarized, for example, by *LeFort and Van der Voo* [1981]), a mechanism for bringing the block to its present relative position exists. Later stages of Alleghenian deformation may have further modified the position of the block by producing the tilting which has been reported for it [*Ellwood and Whitney*, 1980].

Tectonic History of the Elberton-Sparta Crustal Block

A cross section of the study area has been drawn which incorporates the available geological, geochemical, and geophysical information (Figure 9). Presented in Figure 9 are two time slices, the present day (B in Figure 9) with inferred brittle fault motion [e.g., *Griffin*, 1974; *Snoke et al.*, 1980*a*]

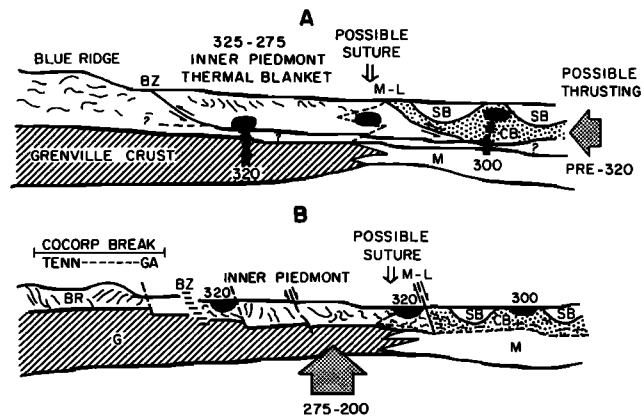


Fig. 9. Cross sections of the study area drawn based on the available geological, geochemical, and geophysical data for the southeastern Appalachian Piedmont. (*a*) Possible suture between North America and the Elberton-Sparta block microplate is illustrated. Pre-320-m.y.-old configuration of the Piedmont, which is intruded by 350- to 250-m.y.-old granites. BZ = Brevard fault zone; M-L = Middleton-Lowndesville fault zone; SB = Slate belt; CB = Charlotte belt, M = mafic crustal material. (*b*) Present configuration of the Piedmont crustal block. Granites intruded 350–200 m.y. (shaded arrow). G = Grenville crust. COCORP break is the data gap between the COCORP lines 1 of Tennessee and Georgia [*Cook et al.*, 1979] (Figure 2). Down to the east, fault surfaces in the Inner Piedmont are based on our mapping at the University of Georgia and on examination of the COCORP line across Georgia.

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which appears to have occurred 275–200 m.y., and a pre-320-m.y. cross section (A in Figure 9) into which has been intruded 320-m.y.-old and younger granites. Interpretation of the data indicates (1) a thrust event which occurred before emplacement of the 320-m.y.-old granites, (2) gentle folding and downwarping of the crust, with the development of a thermal blanket over the Inner Piedmont during the Alleghenian (Hercynian), and (3) rebound of the Inner Piedmont with high erosion rates sometime before 200 m.y. To incorporate the microplate hypothesis into these events, it is necessary to assume that a suture is present slightly to the left of the Elberton granite in Figure 9.

385–415 m.y. Old Granites

RM results after af demagnetization for the Bald Rock (BR), Lowrys (LO), Newberry (NB), Salisbury (SA), Southmont (SO), and Yadkin (YD) granites are reported in Table 2*b*. Only the Lowrys granite exhibits reasonably good within-site and between-site precision, and then for only two sites. A northern hemisphere paleopole calculated for this unit ($\theta = 39.1$; $\phi = 116.2$; a southern paleopole is reported in Table 3) yields a pole, similar to other Middle Paleozoic poles for North America (Figure 7), which is consistent with the 399 ± 4 m.y. old age for the unit [*Fullagar*, 1971] (Table 1). While this result cannot be used to argue for or against an allochthonous mode for the granite, it tentatively indicates that the Lowrys was magnetized as part of the North American continent and was not brought to North America as part of some other continental land mass.

Two of the other 400-m.y.-old granites (the Yadkin and Salisbury granites) exhibited one site each with high within-site precision and similar between-unit directions (Table 2*b*). A paleopole calculated from these directions ($\theta = 73.0$; $\phi = 28.7$) does not correspond to the established polar wander path for North America. It is assumed that the remanent magnetic directions in these granites have been affected by the reported postemplacement faulting and greenschist metamorphism in the area [*Butler and Fullagar*, 1978], while directions for the Lowrys have not.

CONCLUSIONS

The following are concluded from the paleomagnetic investigation of southeast Appalachian Piedmont granites and gneisses reported here:

1. Six granites and gneisses in the vicinity of Atlanta, Georgia, have acquired a characteristic magnetization which is consistent with results determined for other North American rocks of similar age (365–325 m.y.).

2. One 400-m.y.-old granite from northern South Carolina also acquired a characteristic magnetization which is consistent with results determined for other North American rocks of similar age.

3. Granites intruded into the Elberton-Sparta Crustal Block, to the southeast of an arc drawn from Columbia, S. C., through Athens, Ga., to Macon, Ga., ranging in age from 350 to 250 m.y. exhibit a steeper than expected paleomagnetic pole. Two possible explanations for the data from this block are indicated. First, that two episodes of slight magnitude tilting have occurred; one during the Alleghenian event with gentle tilting to the NW as a result of regional warping, and the other as a result of isostatic rebound. This mobility occurred between 320 and 200 m.y. Second, that during the Late Paleozoic this block was accreted as a microplate to North America and may have undergone late stage small scale tilting toward the end of the Alleghenian orogenic event.

4. It is concluded from the available geochemical, geological, and geophysical data obtained for the southern Appalachian Piedmont, that any major thrust event (>50 km) which affected the area must have done so before emplacement of the late-orogenic granites at ~320 m.y. The oxygen and strontium isotopic, gravity, and magnetic susceptibility data are interpreted to indicate that any Alleghenian (Hercynian thrusting) of the Piedmont was of relatively low magnitude (<50 km).

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